

# The Effect of Expected Income on Individual Migration Decisions

John Kennan and James R. Walker

Tate Mason

John Munro Godfrey Sr. Dept. of Economics  
University of Georgia

September 24, 2025

# Roadmap (Talk-First Design)

- ① Motivation & Stylized Facts
- ② Model Intuition
- ③ Estimation Idea
- ④ Results: What Drives Migration?
- ⑤ Interpretation & Policy Angles
- ⑥ Q&A (Appendix: all proofs/derivations/robustness)

# Roadmap

- 1 Motivation & Facts
- 2 Model Intuition
- 3 Estimation Idea
- 4 Results
- 5 Interpretation & Policy
- 6 Conclusion

# Question & Motivation

- **Question:** Do expected income differences drive interstate migration?
- Migration is common for the young; repeat and return moves are pervasive.
- Policy relevance: local shocks, subsidies, welfare differentials.

# Stylized Facts (NLSY79)

TABLE I  
INTERSTATE MIGRATION, NLSY 1979–1994<sup>a</sup>

|                                   | Less Than<br>High School | High<br>School | Some<br>College | College | Total |
|-----------------------------------|--------------------------|----------------|-----------------|---------|-------|
| Number of people                  | 322                      | 919            | 758             | 685     | 2,684 |
| Movers (age 18–27)                | 80                       | 223            | 224             | 341     | 868   |
| Movers (%)                        | 24.8%                    | 24.3%          | 29.6%           | 49.8%   | 32.3% |
| Moves per mover                   | 2.10                     | 1.95           | 1.90            | 2.02    | 1.98  |
| Repeat moves (% of all moves)     | 52.4%                    | 48.7%          | 47.4%           | 50.5%   | 49.5% |
| Return migration (% of all moves) |                          |                |                 |         |       |
| Home                              | 32.7%                    | 33.1%          | 29.1%           | 23.2%   | 28.1% |
| Not home                          | 15.5%                    | 7.1%           | 6.8%            | 8.6%    | 8.4%  |
| Movers who return home            | 61.3%                    | 56.5%          | 51.3%           | 42.8%   | 50.2% |

<sup>a</sup>The sample includes respondents from the cross-section sample of the NLSY79 who were continuously interviewed from ages 18 to 28 and who never served in the military. The home location is the state of residence at age 14.

- Repeat moves are about half of all moves; many are returns to origin.
- Movers are younger; return migration is substantial.

# Roadmap

- 1 Motivation & Facts
- 2 Model Intuition**
- 3 Estimation Idea
- 4 Results
- 5 Interpretation & Policy
- 6 Conclusion

# Model Basics

- **Setup:** Locations differ in mean wages and match quality; moving is costly.
- **Information:** Current location wage known; other locations' matches known only by visiting (experience good).
- **Decision:** Maximize expected lifetime income net of moving costs.

State at  $t$ :

current location  $\ell_0$ , known  $w_{\ell_0}$

prob.  $v(\text{stay})$

prob.  $\mathbb{E}(v(j))$

Stay

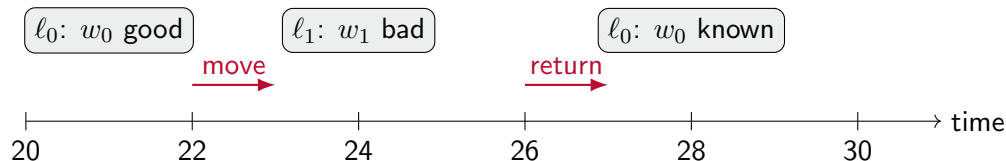
Move to  $j \neq \ell_0$

$$u_t = \alpha_0 w_{\ell_0} + \text{amenities} + \zeta_{\ell_0} \\ + \beta[V_{t+1} | \text{age} + 1, \ell_0]$$

$$u_t = \alpha_0 w_j - \Delta(\ell_0, j) \\ + \text{amenities} + \zeta_j \\ + \beta \mathbb{E}[V_{t+1} | \text{age} + 1, j]$$

## Why Dynamic & Many Locations?

- Return migration and onward moves require dynamics.
- Many alternatives matter: returning is a choice among many states.





# Roadmap

- 1 Motivation & Facts
- 2 Model Intuition
- 3 Estimation Idea**
- 4 Results
- 5 Interpretation & Policy
- 6 Conclusion

## Computational Challenge → Limited-History Approximation

- Full state space explodes with locations and match histories.
- **Idea:** Keep only the most recent  $M$  locations for wages/matches.
- Preserves behavior seen in the data while making DP feasible.

Full state (infeasible):  
(all past locations,  $\{\text{all } w, \xi\}$ )

truncate to last  $M$

Limited history ( $M$  recent):  
 $((\ell_0, \ell_1), (w_{\ell_0}, w_{\ell_1}), (\xi_{\ell_0}, \xi_{\ell_1}))$

## Wage Components (What Matters for Moves)

- Mean wage by location; permanent match component; person FE; transitory shocks.
- Migration choice depends on *differences* in expected income and moving cost, not levels.

### Wage decomposition (relevant pieces in bold)

$$\underbrace{w_{ij}(a, t)}_{\text{observed}} = \underbrace{\mu_j}_{\text{mean by location}} + \underbrace{v_{ij}}_{\text{match}} + G(X_i, a, t) + \eta_i + \varepsilon_{ijt}$$

- Migration choice depends on  $\mu_j + v_{ij}$  and moving cost  $\Delta(\ell_0, j)$ .

# Roadmap

- ① Motivation & Facts
- ② Model Intuition
- ③ Estimation Idea
- ④ Results**
- ⑤ Interpretation & Policy
- ⑥ Conclusion

# What Drives Migration? (Headlines)

- **Expected income matters:** moves respond to geographic mean wages and bad current matches.
- **Costs matter:** distance penalty; strong home premium; cheaper to return.
- **Population size** of destination reduces cost (network/proximity channels).

TABLE II  
INTERSTATE MIGRATION, YOUNG WHITE MEN<sup>a</sup>

|                                            | $\hat{\theta}$ | $\hat{\sigma}_{\theta}$ | $\hat{\theta}$ | $\hat{\sigma}_{\theta}$ | $\hat{\theta}$ | $\hat{\sigma}_{\theta}$ | $\hat{\theta}$ | $\hat{\sigma}_{\theta}$ |
|--------------------------------------------|----------------|-------------------------|----------------|-------------------------|----------------|-------------------------|----------------|-------------------------|
| <i>Utility and cost</i>                    |                |                         |                |                         |                |                         |                |                         |
| Disutility of moving ( $\gamma_0$ )        | 4.790          | 0.565                   | 4.514          | 0.523                   | 4.864          | 0.601                   | 4.851          | 0.604                   |
| Distance ( $\gamma_1$ ) (1000 miles)       | 0.265          | 0.182                   | 0.280          | 0.178                   | 0.311          | 0.187                   | 0.270          | 0.184                   |
| Adjacent location ( $\gamma_2$ )           | 0.808          | 0.214                   | 0.787          | 0.211                   | 0.773          | 0.220                   | 0.804          | 0.216                   |
| Home premium ( $\alpha^H$ )                | 0.331          | 0.041                   | 0.267          | 0.031                   | 0.332          | 0.047                   | 0.337          | 0.045                   |
| Previous location ( $\gamma_3$ )           | 2.757          | 0.356                   | 2.544          | 0.300                   | 3.082          | 0.449                   | 2.818          | 0.416                   |
| Age ( $\gamma_4$ )                         | 0.055          | 0.020                   | 0.062          | 0.019                   | 0.060          | 0.020                   | 0.054          | 0.020                   |
| Population ( $\gamma_5$ ) (millions)       | 0.653          | 0.179                   | 0.653          | 0.178                   | 0.635          | 0.177                   | 0.652          | 0.179                   |
| Stayer probability                         | 0.510          | 0.078                   | 0.520          | 0.079                   | 0.495          | 0.087                   | 0.508          | 0.082                   |
| Cooling ( $\alpha_1$ ) (1000 degree-days)  | 0.055          | 0.019                   | 0.036          | 0.019                   | 0.048          | 0.018                   | 0.056          | 0.019                   |
| Income ( $\alpha_0$ )                      | 0.312          | 0.100                   | —              | —                       | —              | —                       | 0.297          | 0.116                   |
| Location match preference ( $\tau_{\xi}$ ) | —              | —                       | —              | —                       | 0.168          | 0.049                   | 0.070          | 0.099                   |
| <i>Wages</i>                               |                |                         |                |                         |                |                         |                |                         |
| Wage intercept                             | -5.165         | 0.244                   | -5.175         | 0.246                   | -5.175         | 0.246                   | -5.168         | 0.244                   |
| Time trend                                 | -0.035         | 0.008                   | -0.033         | 0.008                   | -0.033         | 0.008                   | -0.035         | 0.008                   |
| Age effect (linear)                        | 7.865          | 0.354                   | 7.876          | 0.356                   | 7.877          | 0.356                   | 7.870          | 0.355                   |
| Age effect (quadratic)                     | -2.364         | 0.129                   | -2.381         | 0.130                   | -2.381         | 0.130                   | -2.367         | 0.129                   |
| Ability (AFQT)                             | 0.014          | 0.065                   | 0.015          | 0.066                   | 0.014          | 0.066                   | 0.014          | 0.065                   |
| Interaction (Age, AFQT)                    | 0.147          | 0.040                   | 0.152          | 0.040                   | 0.152          | 0.040                   | 0.147          | 0.040                   |
| Transient s.d. 1                           | 0.217          | 0.007                   | 0.218          | 0.007                   | 0.218          | 0.007                   | 0.217          | 0.007                   |
| Transient s.d. 2                           | 0.375          | 0.015                   | 0.375          | 0.015                   | 0.375          | 0.015                   | 0.375          | 0.015                   |
| Transient s.d. 3                           | 0.546          | 0.017                   | 0.547          | 0.017                   | 0.547          | 0.017                   | 0.546          | 0.017                   |
| Transient s.d. 4                           | 1.306          | 0.028                   | 1.307          | 0.028                   | 1.307          | 0.028                   | 1.306          | 0.028                   |
| Fixed effect 1                             | 0.113          | 0.035                   | 0.112          | 0.035                   | 0.112          | 0.035                   | 0.113          | 0.035                   |
| Fixed effect 2                             | 0.298          | 0.035                   | 0.296          | 0.035                   | 0.296          | 0.035                   | 0.298          | 0.035                   |
| Fixed effect 3                             | 0.936          | 0.017                   | 0.934          | 0.017                   | 0.934          | 0.017                   | 0.936          | 0.017                   |
| Wage match ( $\tau_v$ )                    | 0.384          | 0.017                   | 0.387          | 0.018                   | 0.387          | 0.018                   | 0.384          | 0.018                   |
| <i>Log likelihood</i>                      |                |                         |                |                         |                |                         |                |                         |
| Exclude income: $\chi^2(1)$                | -4,214.880     |                         | -4,221.426     |                         | -4,218.800     |                         | -4,214.834     |                         |
|                                            |                |                         | 13.09          |                         | 7.93           |                         |                |                         |
| Exclude match preference: $\chi^2(1)$      |                | 0.09                    |                | 5.25                    |                |                         |                |                         |

<sup>a</sup>There are 4,274 (person-year) observations, 432 individuals, and 124 moves.

# Economic Magnitudes

- Moving cost equivalent  $\approx$  \$300k (PV terms, order of magnitude).
- Good match + high-mean state  $\Rightarrow \approx$  \$22k / year gain (1989).

# Tables of Magnitudes

TABLE III  
WAGE PARAMETER ESTIMATES (IN 2010 DOLLARS)

|                      | AFQT Percentile    |                       |                        |
|----------------------|--------------------|-----------------------|------------------------|
|                      | 25                 | 50                    | 75                     |
| <i>Average wages</i> |                    |                       |                        |
| Age 20 in 1979       | 25,827             | 27,522                | 29,216                 |
| Age 20 in 1989       | 18,472             | 20,166                | 21,861                 |
| Age 30 in 1989       | 40,360             | 42,850                | 45,340                 |
|                      | Low                | Middle                | High                   |
| Location match       | −8,366             | 0                     | 8,366                  |
| Fixed effect support | −20,411            | −6,498 −2,454         | 0 2,454 6,498          |
| State means          | Low (WV)<br>12,698 | Rank 5 (OK)<br>14,530 | Median (MO)<br>16,978  |
|                      |                    |                       | Rank 45 (RI)<br>19,276 |
|                      |                    |                       | High (MD)<br>22,229    |

TABLE IV  
MOVING COST EXAMPLES

|               | $\gamma_0$ | $\alpha_0$ | Age    | Distance | Adjacent | Population | Previous | Cost      |
|---------------|------------|------------|--------|----------|----------|------------|----------|-----------|
| $\theta$      | 4.790      | 0.312      | 0.055  | 0.265    | 0.808    | 0.653      | 2.757    |           |
| Young mover   |            |            | 20     | 1        | 0        | 1          | 0        | \$384,743 |
| Average mover |            |            | 24.355 | 0.664    | 0.427    | 0.728      | 0.371    | \$312,146 |

# Fit & Mechanism

- Model reproduces repeat/return migration shares qualitatively.
- Negative association: higher realized income this period  $\Rightarrow$  lower out-migration probability.



# Roadmap

- 1 Motivation & Facts
- 2 Model Intuition
- 3 Estimation Idea
- 4 Results
- 5 Interpretation & Policy**
- 6 Conclusion

# What We Learn

- Migration resembles optimal search over locations.
- Both mean differences and idiosyncratic match quality drive moves.
- Structural approach enables policy counterfactuals.

# Policy Angles & Extensions

- Local labor demand shocks & wage equalization dynamics.
- Subsidies or relocation assistance: who moves more?
- Household joint decisions; welfare benefit differentials; retirement migration.

# Roadmap

- ① Motivation & Facts
- ② Model Intuition
- ③ Estimation Idea
- ④ Results
- ⑤ Interpretation & Policy
- ⑥ Conclusion

# Takeaways

- Expected income plays a sizable role in interstate migration.
- High moving costs + payoff shocks reconcile low average move rates with meaningful gains.
- Model provides a tractable template for counterfactual policy analysis.

# Thanks!

Questions? [tate.mason@uga.edu](mailto:tate.mason@uga.edu)

# Roadmap

## 7 Appendix

## Data & Sample Construction (*Appendix*)

- NLSY79 cross-section; white non-Hispanic HS grads; ages 20+; not in military.
- Attrition and consistency checks; AFQT use; state means from Census PUMS (1990).



# Sample Selection

|                                                      | Respondents | Person-Years |
|------------------------------------------------------|-------------|--------------|
| White non-Hispanic males (cross-section sample)      | 2,439       | 39,024       |
| <i>Restrictions applied to respondents</i>           |             |              |
| Ever in military                                     | −246        |              |
| High school dropouts and college graduates           | −1,290      |              |
| Attended college                                     | −130        |              |
| Older than age 20 at start of sample period          | −134        |              |
| Missing AFQT score                                   | −41         |              |
| Attend or graduate from high school at age 20        | −87         |              |
| Not in labor force for more than 1 year after age 19 | −44         |              |
| Location at age 20 not reported                      | −20         |              |
| Income information inconsistent                      | −1          |              |
| Died before age 30                                   | −4          |              |
| Residence at age 14 not reported                     | −2          |              |
| In jail in 1993                                      | −1          |              |
| Subtotal                                             | −2,000      | 6,585        |
| <i>Restrictions applied to periods</i>               |             |              |
| Delete periods after first gap in history            | −1          | −1,104       |
| Delete periods before age 20                         | −6          | −1,207       |
| <b>Analysis sample</b>                               | <b>432</b>  | <b>4,274</b> |

# Sample Selection — Years per person

|                         | Respondents | Person-Years |
|-------------------------|-------------|--------------|
| <i>Years per person</i> |             |              |
| 1                       | 14          | 14           |
| 2                       | 16          | 32           |
| 3                       | 19          | 57           |
| 4                       | 14          | 56           |
| 5                       | 14          | 70           |
| 6                       | 14          | 84           |
| 7                       | 13          | 91           |
| 8                       | 9           | 72           |
| 9                       | 34          | 306          |
| 10                      | 61          | 610          |
| 11                      | 53          | 583          |
| 12                      | 44          | 528          |
| 13                      | 45          | 585          |
| 14                      | 44          | 616          |
| 15                      | 38          | 570          |
| <b>Total</b>            | <b>432</b>  | <b>4,274</b> |

# Value Function & Choice Probabilities (*Appendix*)

## Bellman equation

$$V(x, \zeta) = \max_{j \in \mathcal{J}} \{ v(x, j) + \zeta_j \},$$
$$v(x, j) = u(x, j) + \beta \mathbb{E}_{x'} [\bar{V}(x') \mid x, j], \quad \bar{V}(x) = \mathbb{E}_{\zeta} [V(x, \zeta)].$$

## Extreme value shocks $\Rightarrow$ logit choice probabilities

$$\rho(x, j) = \frac{\exp(v(x, j))}{\sum_{k \in \mathcal{J}} \exp(v(x, k))}, \quad \exp(\bar{V}(x)) = e^{\bar{\gamma}} \sum_k \exp(v(x, k)).$$

# Limited-History State Transitions (*Appendix*)

## State and transitions (with $M = 2$ recent locations)

State:  $x = (\ell_0, \ell_1, \omega_{\ell_0}^{(w)}, \omega_{\ell_1}^{(w)}, \omega_{\ell_0}^{(\xi)}, \omega_{\ell_1}^{(\xi)}, a)$ .

If stay:  $x' = (\ell_0, \ell_1, \cdot, \cdot, \cdot, \cdot, a + 1)$ .

If move to previous  $\ell_1$ : swap  $(\ell_0, \ell_1)$  and their components,  $a \leftarrow a + 1$ .

If move to new  $j$ : shift  $(\ell_0 \leftarrow j, \ell_1 \leftarrow \ell_0)$  and draw new  $(\omega_j^{(w)}, \omega_j^{(\xi)})$ .

## Likelihood & Identification Sketch (*Appendix*)

- Joint likelihood of income & choices; mixture over unobserved components.
- Nonparametric intuition: selection on bad current draws  $\uparrow$  move probability.

## Robustness (*Appendix*)

- Vary  $\beta$ , horizon  $T$ , include preference match, extra amenities.
- Fit diagnostics; simulated vs. observed migration rates.

# Robustness (Appendix)

TABLE XIV  
ESTIMATES FROM SIMULATED MIGRATION HISTORIES

|                              | Base Model     |                         | 100 Reps       |                         | <i>t</i> |
|------------------------------|----------------|-------------------------|----------------|-------------------------|----------|
|                              | $\hat{\theta}$ | $\hat{\sigma}_{\theta}$ | $\hat{\theta}$ | $\hat{\sigma}_{\theta}$ |          |
| Disutility of moving         | 4.790          | 0.565                   | 4.732          | 0.058                   | -1.00    |
| Distance                     | 0.265          | 0.182                   | 0.272          | 0.015                   | 0.42     |
| Adjacent location            | 0.808          | 0.214                   | 0.808          | 0.017                   | 0.04     |
| Home premium                 | 0.331          | 0.041                   | 0.333          | 0.004                   | 0.42     |
| Previous location            | 2.757          | 0.356                   | 2.709          | 0.031                   | -1.52    |
| Age                          | 0.055          | 0.020                   | 0.056          | 0.002                   | 0.25     |
| Population                   | 0.653          | 0.179                   | 0.648          | 0.017                   | -0.30    |
| Stayer probability           | 0.510          | 0.078                   | 0.517          | 0.008                   | 1.00     |
| Cooling                      | 0.055          | 0.019                   | 0.055          | 0.002                   | -0.07    |
| Income                       | 0.312          | 0.100                   | 0.307          | 0.008                   | -0.56    |
| Wage intercept               | -5.165         | 0.244                   | -5.158         | 0.033                   | 0.21     |
| Time trend                   | -0.035         | 0.008                   | -0.035         | 0.001                   | 0.40     |
| Age effect (linear)          | 7.865          | 0.354                   | 7.853          | 0.050                   | -0.23    |
| Age effect (quadratic)       | -2.364         | 0.129                   | -2.359         | 0.018                   | 0.28     |
| Ability (AFQT)               | 0.014          | 0.065                   | 0.017          | 0.010                   | 0.35     |
| Interaction (age, AFQT)      | 0.147          | 0.040                   | 0.142          | 0.007                   | -0.67    |
| Transient s.d. 1             | 0.217          | 0.007                   | 0.217          | 0.001                   | 0.07     |
| Transient s.d. 2             | 0.375          | 0.015                   | 0.376          | 0.002                   | 0.37     |
| Transient s.d. 3             | 0.546          | 0.017                   | 0.545          | 0.002                   | -0.84    |
| Transient s.d. 4             | 1.306          | 0.028                   | 1.306          | 0.004                   | 0.14     |
| Fixed effect 1               | 0.113          | 0.035                   | 0.114          | 0.003                   | 0.40     |
| Fixed effect 2               | 0.298          | 0.035                   | 0.300          | 0.003                   | 0.68     |
| Fixed effect 3               | 0.936          | 0.017                   | 0.938          | 0.002                   | 1.37     |
| Wage match                   | 0.384          | 0.017                   | 0.383          | 0.002                   | -0.17    |
| Log likelihood, $\chi^2(24)$ | -4,214.88      |                         | -473,830.3     |                         | 9.95     |